BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2025-2026

SCHOOL OF COMPUTER APPLICATIONS

Assignment

Data Science

(BCADSN14202)

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Submitted To:-

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ASSIGNMENT

Title:

- **Accessing IBM Cloud, Watson Studio and Creating a Jupyter Notebook.**

Objective:

- The objective of this practical is to understand how to access IBM Cloud and use Watson Studio for data science tasks. This experiment aims to create a project in Watson Studio and successfully launch a Jupyter Notebook. It also helps in learning how cloud platforms support data analysis and machine learning workflows.

- This practical helps in gaining hands-on experience with cloud computing tools used in data science.

Assignment Description

- This assignment is based on accessing IBM Cloud and using Watson Studio to create a project and launch a Jupyter Notebook. It helps in understanding how cloud-based platforms are used for data science activities. Screenshots with brief explanations are included to explain each step clearly.

Tools & Technologies Used

- **IBM Cloud Platform –** Used to access cloud services and resources.
- **Watson Studio** – Used for managing data science projects and tools.
- **Jupyter Notebook** – Used for writing and executing Python code interactively.
- **Web Browser (Google Chrome / Edge)** – Used to access IBM Cloud services.
- **Microsoft Word** – Used for documenting the practical with screenshots and explanations.

Problem Statement

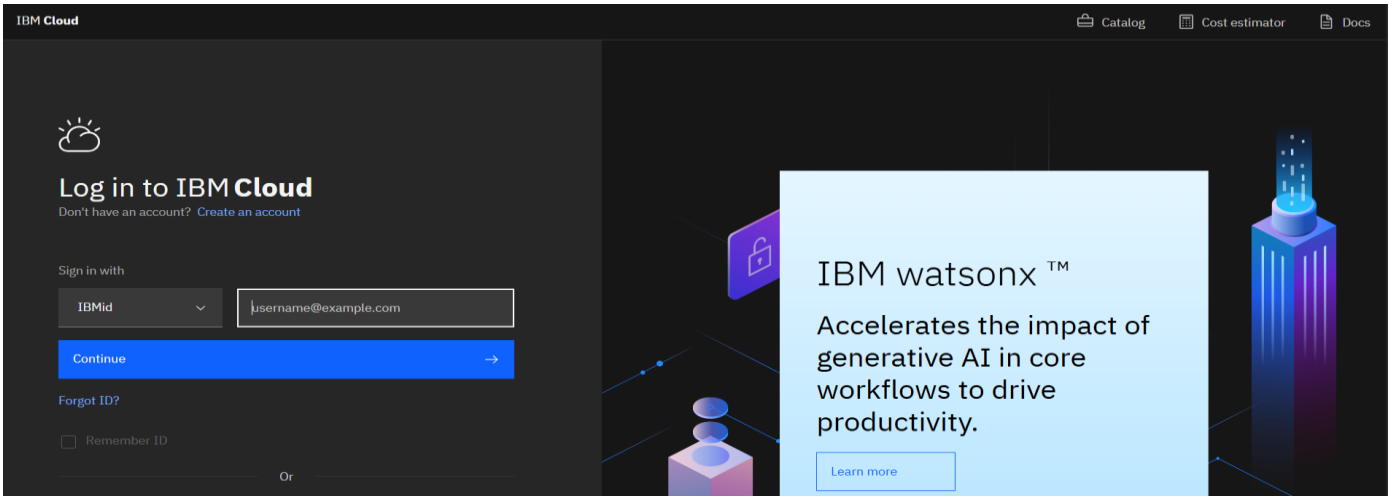
- To log in to IBM Cloud using valid credentials, access Watson Studio, create a new project, and add a Jupyter Notebook inside the project. The task also includes documenting each step with screenshots and brief explanations to understand the workflow of cloud-based data science platforms.

ASSIGNMENT

Question: - How can IBM Cloud and Watson Studio be used to create and run a Jupyter Notebook for data science tasks?

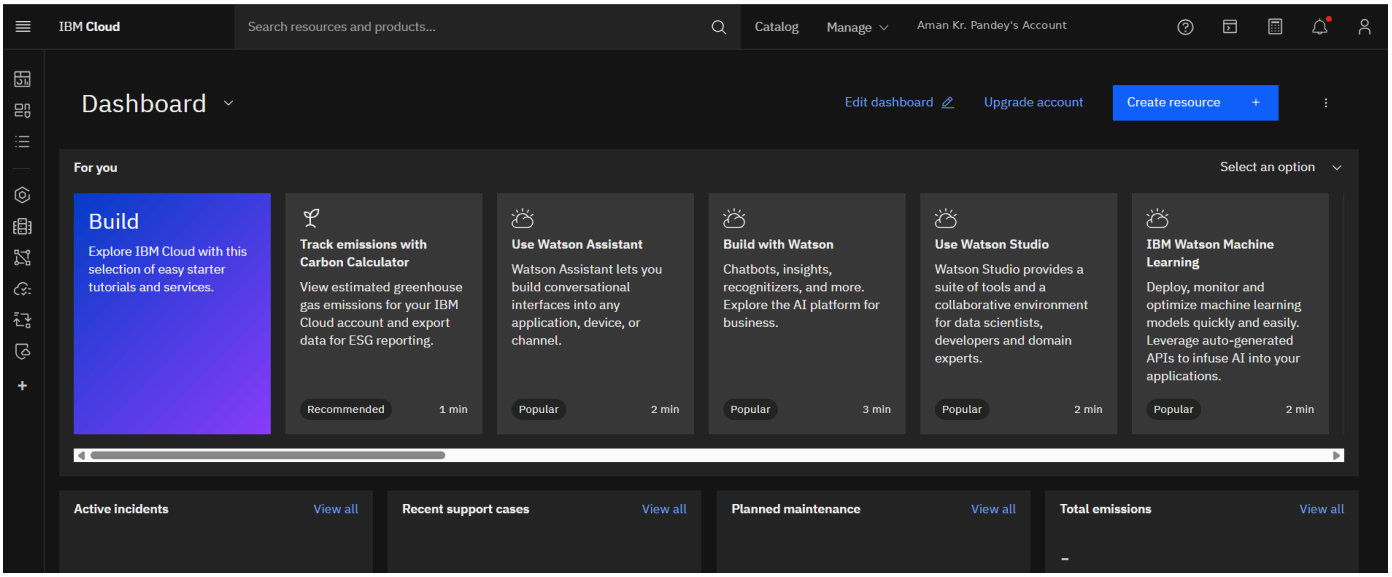
Step 1: - IBM Cloud Website Login

- Open your browser.
- Go to https://cloud.ibm.com/login.



- **Explanation: -** This screenshot shows the IBM Cloud official website login page. Here, the user enters their IBM credentials (email ID and password) to securely access the IBM Cloud platform. Successful authentication allows access to cloud services and resources.

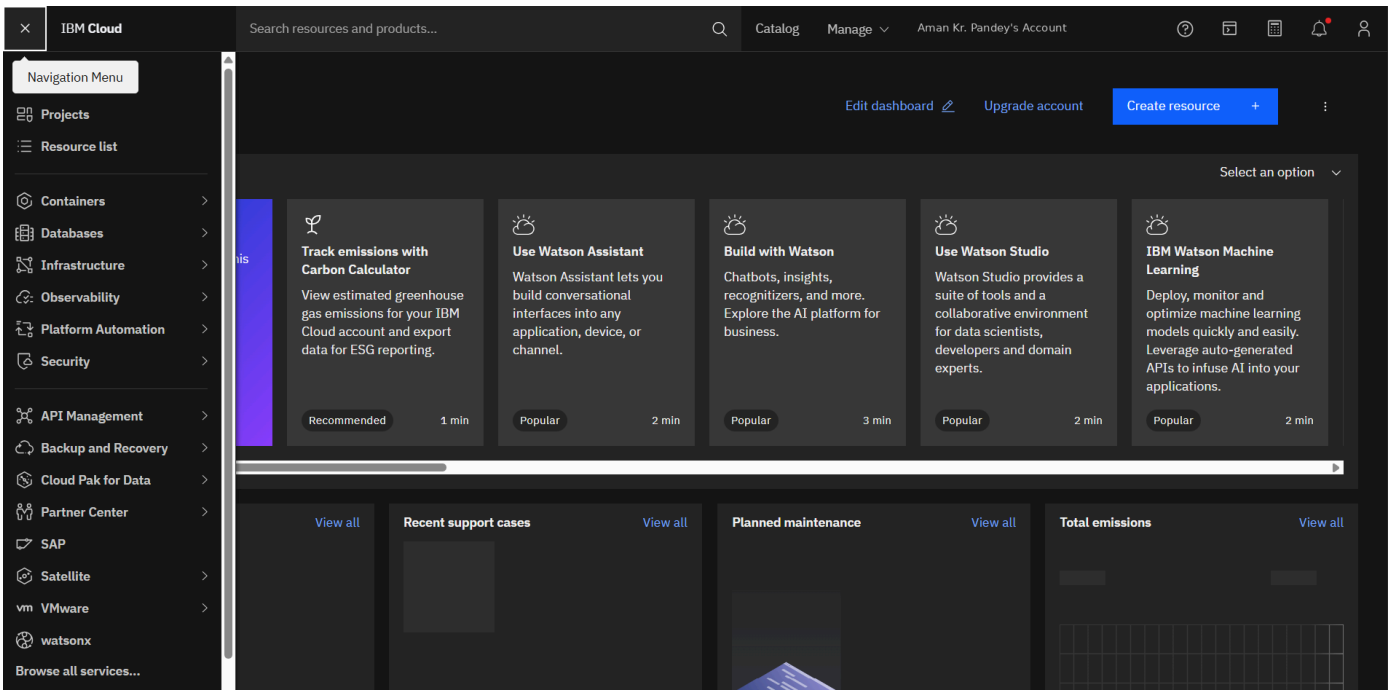
Step 2: - IBM Cloud Dashboard



- **Explanation: -** After successful login, the IBM Cloud dashboard is displayed. This dashboard provides an overview of services, resources, and navigation options available to the user. From here, different cloud tools can be accessed.

Step 3: - Accessing Watson Studio

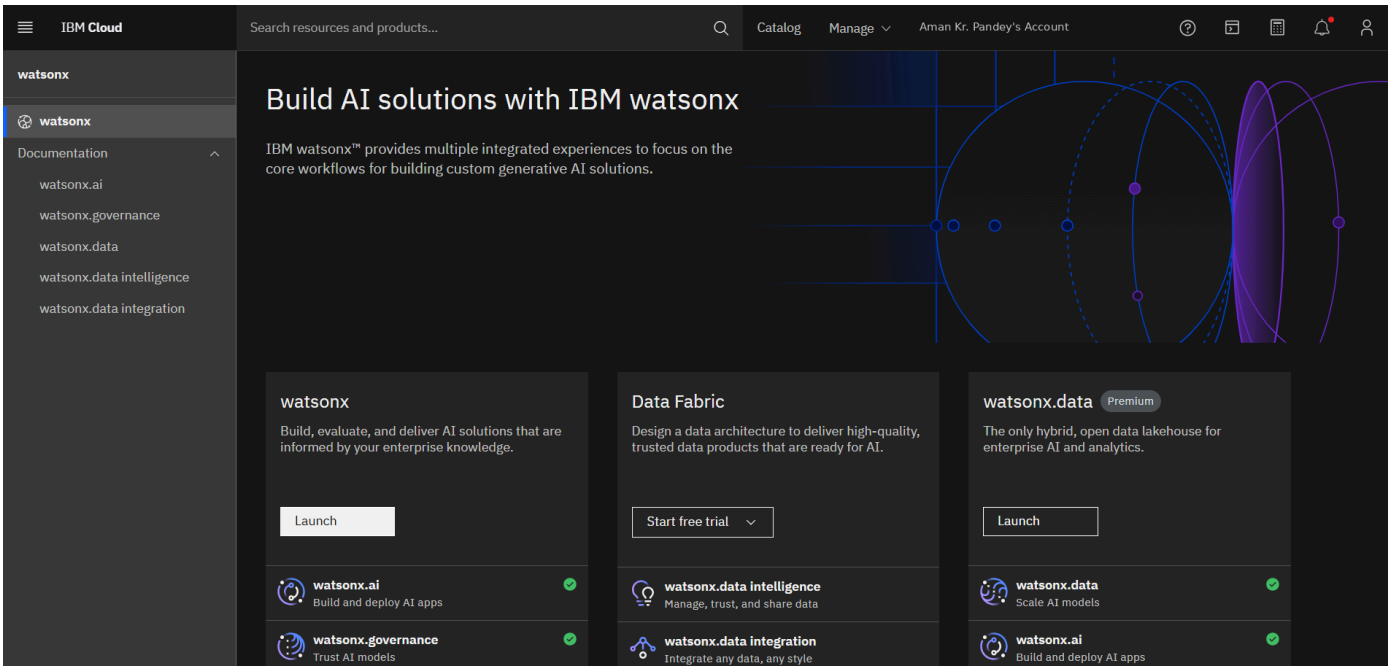
- After opening the official website of **"IBM Cloud official site"**, from the left and top corners of the page, click the navigation menu icon.



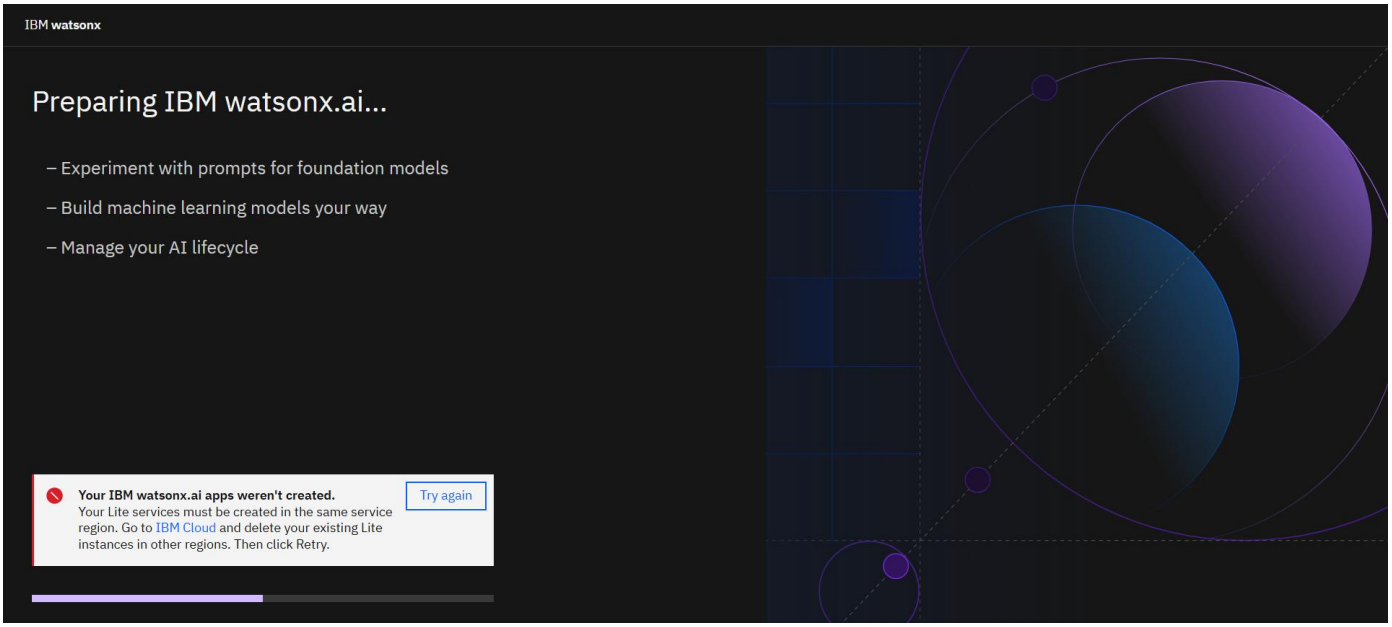
- **Explanation: -** This screenshot shows how Watson Studio is accessed from the IBM Cloud dashboard. After that, click **"watsonx"** to access Watson Studio. Watson Studio is a data science platform used for building, training, and managing machine learning models and notebooks.

Step 4: - Watson Studio Home Page

- After clicking on **"watsonx"**, the display looks like this.
- From this page, click on the **"Launch"** button on **"watsonx".**



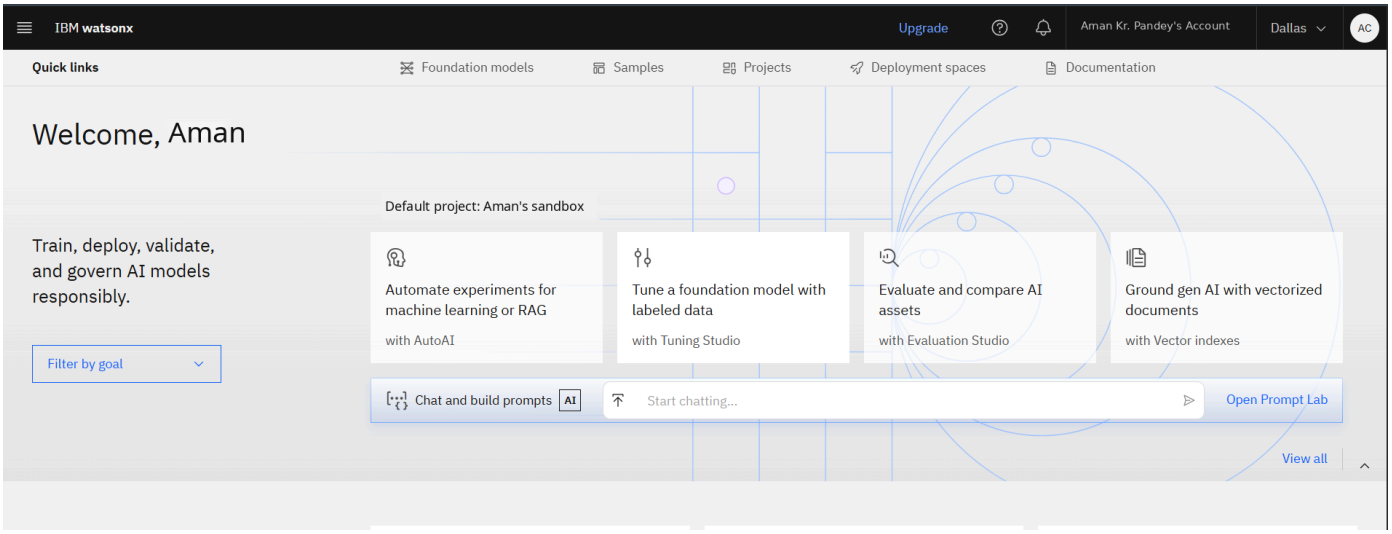
- After clicking on **"Launch",** you look at this page.
- Click on **"IBM Cloud"**.



- **Explanation: -** As soon as you click on **"IBM Cloud"**, the Resource List page will open. On that page, all Watsonx services listed there need to be deleted one by one.

  After deleting the resources, return to the same page and click on **"Try Again."**

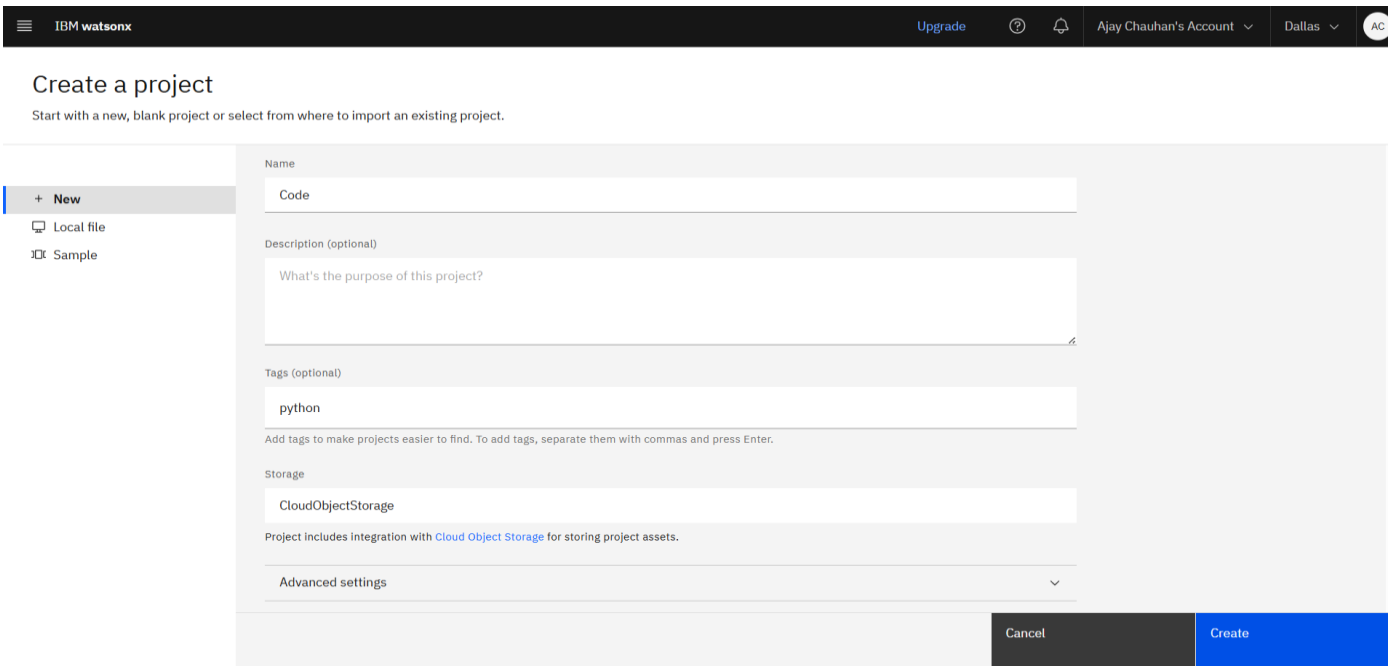
- Once you do that, the **"official Watson Studio page"** will open.



- **Explanation: -** This screenshot displays the Watson Studio home page. It provides options to create new projects, manage existing projects, and access data science tools such as "**Jupyter Notebooks**" and "**machine learning services**".

Step 5: - Creating a New Project

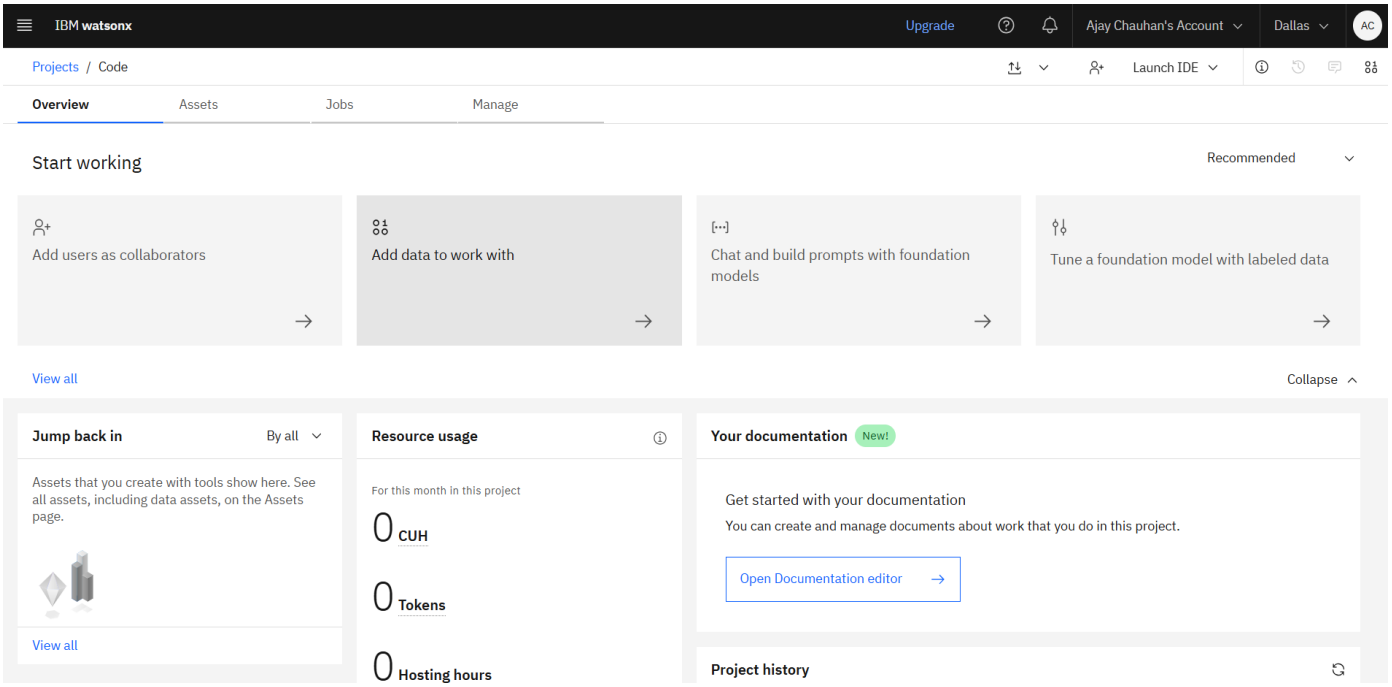
- After that clicking on **"Project"** → **"New Project".**



- **Explanation: -** This screenshot shows the project creation process, where the project name and storage details are provided. After filling in the required information and clicking on **"Create"**, the project is successfully created and is ready for use.

Step 6: - Project Dashboard

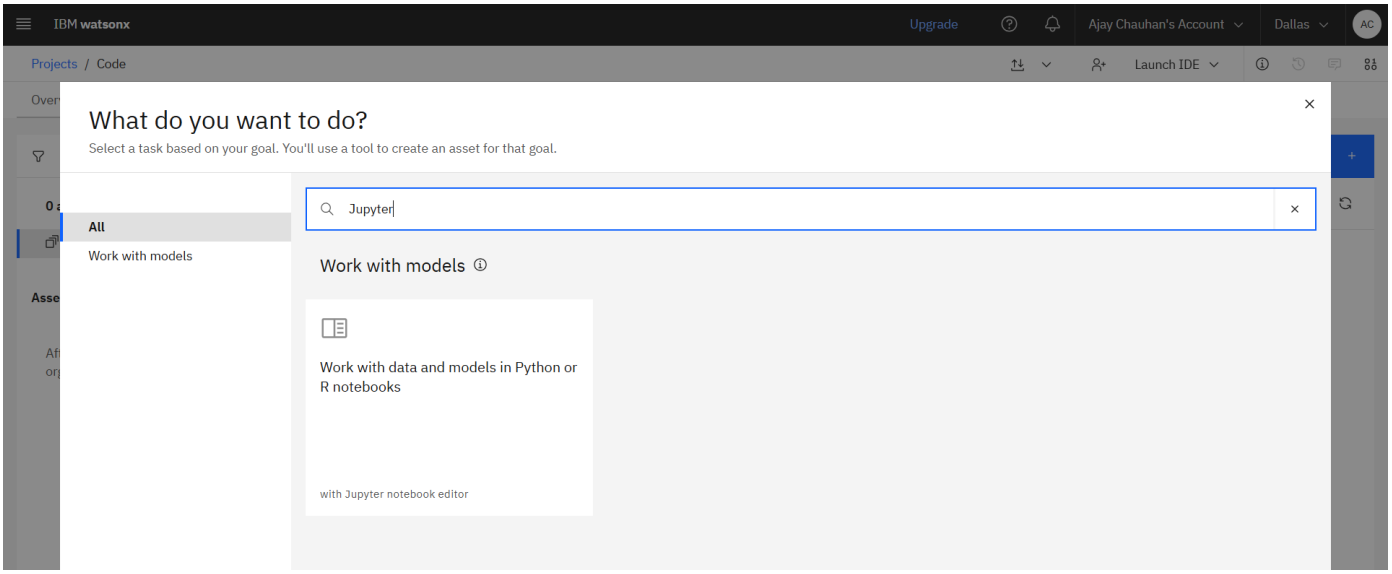
- From this page, click on **"Assets"** beside **"Overview."**



- **Explanation: -** This screenshot shows the dashboard of the newly created project. From here, users can add assets such as **"Jupyter Notebooks"**, **"data files"**, and models for data analysis tasks.

Step 7: - Adding a Jupyter Notebook

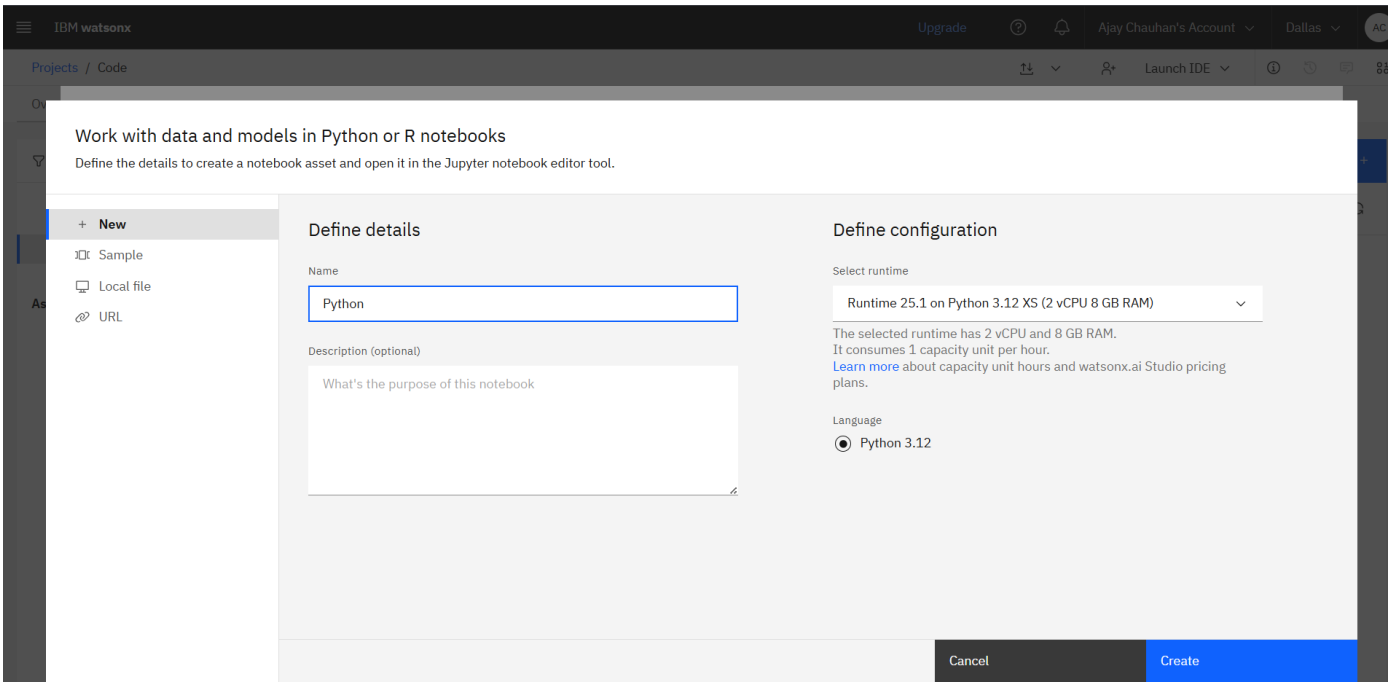
- search and click on "with Jupyter Notebook editor".



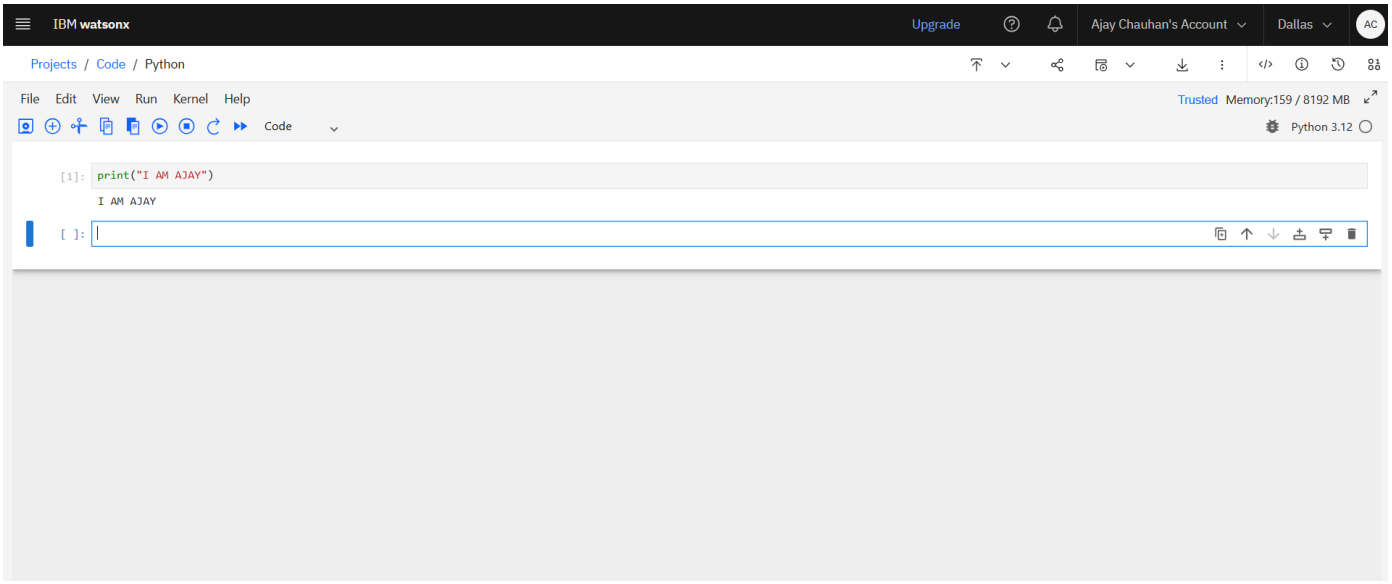
- **Explanation: -** This screenshot shows the option to search and add a **"Jupyter Notebook"** inside the project. **"Jupyter Notebook"** is an interactive an environment used for writing and executing Python code for data analysis and machine learning.

Step 8: - Jupyter Notebook Interface

- After clicking on **"Jupyter Notebook"**, fill in the required details like:- (Name, description, selected runtime) and then click on **"Create."**



➢ After clicking on **"Create"**, the page will open as shown below.



❖ **Explanation: -** This screenshot displays the "**Jupyter Notebook interface"** after it opens successfully. Users can write code, add Markdown text, run cells, and visualise outputs directly within **Watson Studio**.